Math 242, S20 **Name:** Answer

Quiz 7

__Instructions:__ **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (5 points) Find the solution of the following initial value problem:

$$y'' + 8y' + 25y = 0, \quad y(0) = 4, \quad y'(0) = -1.$$

Characteristic equation.

$$r^2 + 8r + 25 = 0$$

$$r = \frac{-8 \pm \sqrt{64 - 100}}{2} \quad \Rightarrow y(x) = e^{-4x}(C_1 \cos 3x + C_2 \sin 3x)$$

$$= -4 \pm 3i \quad y'(x) = e^{-4x}(-3C_1 \sin 3x + 3C_2 \cos 3x) - 4e^{-4x}(C_1 \cos 3x + C_2 \sin 3x)$$

$$y(0) = 1 \Rightarrow e^0(C_1 \cdot 1 + C_2 \cdot 0) = 4$$

$$\Rightarrow C_1 = 4$$

$$y'(0) = 1 \Rightarrow e^0(3C_2) - 4e^0(C_1) = -1$$

$$\Rightarrow -4C_1 + 3C_2 = -1 \Rightarrow C_2 = \frac{1}{3}(4C_1 - 1) = \frac{1}{3}(16 - 1) = 5$$

thus $y(x) = e^{-4x}(4\cos 3x + 5\sin 3x)$

Problem 2. (5 points) Find the general solution of the homogeneous differential equations

$$4y^{(3)} + 12y'' + 9y' = 0.$$

Characteristic equation:

$$4r^3 + 12r^2 + 9r = 0$$

$$r(4r^2 + 12r + 9) = 0$$

$$r(2r + 3)^2 = 0$$

$$\Rightarrow r_1 = 0, \quad r_2 = r_3 = -\frac{3}{2}$$

Thus $y(x) = C_1 + (C_2 + C_3 x)e^{-\frac{3}{2}x}$